SPLITTING OF PURE BRAID MONODROMY FOR THE DOUBLE-BUBBLE SYMPLECTIC PLUMBING

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Allowed standard prerequisites (editorial).

- Autoequivalence groups and connected stability spaces of type A Calabi-Yau categories. Ikeda Theorems7.15,7.16; Qiu-Woolf Theorems B; BS Theorem9.9, all cited precisely in author proof
- Cluster categories and exchange-graph quotient. Baur-Marsh, BRT, King-QiuYu-cluster Theorem8.6
- Koszul duality and Rabinowitz/cluster equivalence for Dynkin plumbings. AbSm-plumbing, Kalck-Yang, BJK Corollary1.4 and Lekili-Ueda as explicitly cited

Source reference. Original section 7: algebra-geometry/sources/sigma-2026-080/native/sigma26-080.tex, source lines 1307–1307. Forward pointer in introductory categorical scope only, unrelated to the type A splitting proof.

Source reference. Original reference 6: algebra-geometry/sources/sigma-2026-080/native/sigma26-080.tex, source lines 1244–1246. Source narrative comparison to another independent geometric case only; none of its results is used as an input to the retained proof.

1. Flavours of Fukaya category

Let (X, ω) be a connected symplectic manifold with $2c_1(X) = 0$. We will fix a Maslov cover of the Lagrangian Grassmannian, meaning that there is a well-defined group $\mathrm{Symp}^{\mathrm{gr}}(X)$ of graded symplectomorphisms, and a subgroup $\mathbb{Z} \leqslant \mathrm{Symp}^{\mathrm{gr}}(X)$ generated by the shift (i.e., the fibrewise deck transformation). Recall from [74, Lemma 2.4] the following.

Lemma 1.1. *There is an exact sequence*

$$1 \to \mathbb{Z} \to \mathrm{Symp}^{\mathrm{gr}}(X) \to \mathrm{Symp}(X) \to H^1(X; \mathbb{R}).$$

The final map is not a homomorphism but its kernel is a subgroup.

Lemma 1.2. *Suppose that X is non-compact. The forgetful map* $\mathrm{Symp}^{\mathrm{gr}}(X) \to \mathrm{Symp}(X)$ *is split over the subgroup* $\mathrm{Symp}_{\mathrm{ct}}(X)$.

Proof. Take the unique gradings for symplectomorphisms which act trivially on the fibres of the Maslov cover near infinity. □

Let (X, ω) be a Liouville manifold, with contact type boundary, and assume $2c_1(X) = 0$. We will be interested in all of

(1) the compact Fukaya category $\mathcal{F}(X)$, with objects compact graded exact spin Lagrangians;
(2) the wrapped Fukaya category $\mathcal{W}(X)$, with objects graded exact spin Lagrangians which are cones over Legendrian submanifolds near infinity;

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(3) the Rabinowitz Fukaya category $\mathcal{R}(X) := \mathcal{W}(X)/\mathcal{W}(X)^{\mathrm{prop}}$, which is the Verdier quotient of the (Yoneda image of the) wrapped category by its subcategory of proper modules.

All three categories are defined over $\mathbb{Z}$, and $\mathcal{F}(X) \subset \mathcal{W}(X)$ is a full and faithful subcategory. We say that 'Koszul duality' holds for the compact and wrapped categories if $\mathcal{F}(X) \subset \mathcal{W}(X)$ is a Koszul dualising subcategory in the sense of [26, Definition 2.12]. This often holds for Weinstein manifolds obtained from plumbings of simply connected pieces. Concretely, the wrapped category of a Weinstein manifold admits a finite set of generators $\{T_i \mid 1 \leqslant i \leqslant N\}$ and one seeks a finite set of compact branes $\{L_i \mid 1 \leqslant i \leqslant N\}$ such that the total endomorphism algebras $\mathcal{Q} = \bigoplus_{i,j} HF^*(L_i, L_j)$ and $\mathcal{A} = \bigoplus_{i,j} HW^*(T_i, T_j)$, which are augmented over the semisimple ring $\Bbbk = \bigoplus_{i=1}^N \mathbb{K}_i$ generated by the units of the objects, are A_∞-Koszul dual, meaning

$$\mathcal{Q} \simeq \mathrm{Ext}_{\mathcal{A}}(\Bbbk, \Bbbk), \qquad \mathcal{A} \simeq \mathrm{Ext}_{\mathcal{Q}}(\Bbbk, \Bbbk).$$

In this setting, there is an equivalence [26, Corollary 1.4]

$$\mathcal{R}(X) \simeq \mathcal{W}(X)/\mathcal{F}(X).$$

This is the formulation we will encounter in practise, but the general algebraic construction of $\mathcal{R}(X)$ from $\mathcal{W}(X)$ in particular shows that autoequivalences of the latter act on the former. (In Section 7, it will be important to work over $\mathbb{C}$, to incorporate $\mathbb{C}^*$-valued local systems on Lagrangians.)

Any autoequivalence of $\mathcal{W}(X)$ arising from a symplectomorphism of X will preserve the compact category $\mathcal{F}(X)$, and hence induce an equivalence of $\mathcal{R}(X)$. For background on $\mathcal{F}(X)$, see [76], and for background on $\mathcal{W}(X)$, see [1]; for background on derived localisation in general, see [22], and in the context of $\mathcal{R}(X)$ specifically, see [26].

The Grothendieck group $K(\mathcal{F}(X))$ is the free abelian group on isomorphism classes of objects, modulo relations coming from exact triangles. For later use, we define the 'numerical Grothendieck group'

$$K_{\mathrm{num}}(\mathcal{F}(X)) \tag{1.1}$$

to be the quotient of $K(\mathcal{F}(X))$ by the subgroup which is the radical of the Euler pairing, i.e., that generated by compact exact Lagrangians $L \in \mathrm{Ob}\,\mathcal{F}(X)$ for which $\chi(HF^*(L, \bullet)) = 0$ for every $\bullet \in \mathcal{W}(X)$.

We will be interested in actions of symplectomorphism groups on Fukaya categories. Especially for actions on the wrapped category, one needs some control of maps at infinity to ensure that (perhaps after isotopy) they preserve the class of Lagrangians which are conical[1] at infinity. We say a symplectomorphism ϕ of a Liouville manifold $(X, \mathrm{d}\theta)$ is *exact* if $\phi^*\theta = \theta + \mathrm{d}f$ for some $f\colon X \to \mathbb{R}$, and *strongly exact* if one can take f to have compact support. Both notions have obvious graded analogues. It is classical that strongly exact maps act on the wrapped category. Write $\mathrm{Symp}_{\mathrm{ex}}(X)$ for the group of exact symplectomorphisms.

Lemma 1.3. *Assume that the Liouville manifold X has finite type, so can be written as the completion of a Liouville domain $\mathring{X}$. Then*

(1) *There is a homomorphism $\pi_0 \mathrm{Symp}_{\mathrm{ct}}(X) \to \pi_0 \mathrm{Symp}_{\mathrm{ex}}(\mathring{X})$;*
(2) *The group $\pi_0 \mathrm{Symp}(\mathring{X})$ acts on both $\mathcal{F}(X)$ and $\mathcal{W}(X)$.*

Proof. For the first statement, conjugation by a suitably large Liouville flow will bring the support into $\mathring{X}$. We claim we can then deform to obtain an exact representative of the map. Let $\omega = \mathrm{d}\lambda$ be the Liouville form on X, and suppose we are

[1]Or more generally geometrically bounded for a dissipative almost complex structure in the sense of Groman [29].

given $\psi \in \mathrm{Symp}_{\mathrm{ct}}(X)$. We have $\psi^*\lambda - \lambda = \theta$, where θ is a closed one-form with compact support. Let V be the vector field which is symplectically dual to $-\theta$. This has compact support, and so is integrable for all time; let σ_t denote its flow. Then applying Cartan's formula, we get that $\psi \circ \sigma_1$ is an exact symplectomorphism. (Note that the proof of [11, Lemma 1.1] specialises to this argument in this case.)

This construction can be performed in families, which shows that the inclusion

$$\mathrm{Symp}_{\mathrm{ct,ex}}(X) \hookrightarrow \mathrm{Symp}_{\mathrm{ct}}(X)$$

is a homotopy equivalence, so induces an isomorphism on π_0. On the other hand, there is then a forgetful map

$$\mathrm{Symp}_{\mathrm{ct,ex}}(X) \to \mathrm{Symp}_{\mathrm{ex}}(\mathring{X})$$

which induces a homomorphism on π_0. Combining these two gives the first statement.

The second statement is exactly [45, Corollary 2.4] (cf. [46], which in particular involves deforming a map of the Liouville domain to be exact, using a version of the argument given in the previous paragraphs). □

Lemma 1.4. *In the setting of Lemma* 1.3*, assume moreover that* $2c_1(\mathring{X}) = 0$*. Then there are compatible representations*

$$\begin{array}{ccc} \pi_0 \mathrm{Symp}^{\mathrm{gr}}_{\mathrm{ct}}(X) & \longrightarrow & \mathrm{Auteq}(\mathcal{F}(X)) \\ \downarrow & & \uparrow \\ \pi_0 \mathrm{Symp}^{\mathrm{gr}}(\mathring{X}) & \longrightarrow & \mathrm{Auteq}(\mathcal{W}(X)). \end{array}$$

The top line is classical; the bottom line is [45, Corollary 2.9] (see also [46]).

Although the Rabinowitz category is not a priori determined by the contact boundary of X, one still has the following lemma.

Lemma 1.5. *The group* $\pi_0 \mathrm{Symp}^{\mathrm{gr}}_{\mathrm{ct}}(X)$ *acts trivially on the Rabinowitz category.*

Proof. Let L, K be Lagrangians which are cones over Legendrian submanifolds L_∞ and K_∞. The morphism group $RFC^*(L,K)$ in $\mathcal{R}(X)$ between L and K is the chain-level mapping cone of a natural continuation map $CW_*(L,K) \to CW^*(L,K)$ coming from a monotone homotopy of Hamiltonians from 'infinite negative wrapping to infinite positive wrapping'. This factors as a composition

$$CW_*(L,K) \xrightarrow[a]{} CF^*(L, \phi_{-\varepsilon}K) \xrightarrow[c]{} CF^*(L, \phi_{+\varepsilon}K) \xrightarrow[b]{} CW^*(L,K),$$

where a, b are projection respectively inclusion maps of summands into the relevant direct systems and c is the natural continuation from small negative to small positive wrapping. Viewing $RFC^*(L,K) = \mathrm{Cone}(b \circ c \circ a)$, there are exact triangles of mapping cones

$$\mathrm{Cone}(c \circ a) \to RFC^*(L,K) \to \mathrm{Cone}(b)$$

and

$$\mathrm{Cone}(a) \to \mathrm{Cone}(c \circ a) \to \mathrm{Cone}(c).$$

The continuation map c induces the Poincaré duality isomorphism in Lagrangian Floer theory

$$HF^*(L, \phi_{-\varepsilon}K) \to HF^*(L, \phi_{+\varepsilon}(K))$$

(which recovers the isomorphism $H_*(L, \partial_\infty L) \to H^{n-*}(L)$ when $L = K$), so has acyclic cone. On the other hand, $\mathrm{Cone}(b) \simeq CW(L,K)^+$ defines the Lagrangian Floer analogue of 'positive symplectic cohomology' [70]; it can be computed by an action truncated version of the wrapped Floer complex which removes action zero chords (i.e., interior intersection points), yielding a complex only built from

Reeb chords at infinity between the Legendrian boundaries. Similarly for Cone(a). Putting these ingredients together, one obtains a cochain complex quasi-isomorphic to that underlying $RFC^*(L,K)$ and which is generated only by Reeb chords at the boundary. Any compactly supported map then acts trivially on the underlying chain group.

Since L and K were arbitrary, compactly supported maps act trivially on the whole A_∞-category. □

To conclude this section, we briefly recall the categories associated to quivers with potential; a more extensive discussion can be found in [83].

A quiver Q is a directed graph, and a potential W on Q is a formal sum of cycles in the edges. To such (Q,W) one can associate a 3-dimensional[2] Calabi–Yau category $\mathcal{C}(Q,W)$ via the 'Ginzburg construction' [28], which defines a non-positively graded dg-algebra $\mathcal{J}(Q,W)$ with $H^0(\mathcal{J}(Q,W)) = J(Q,W)$ the Jacobi algebra (i.e., the path algebra modulo cyclic derivatives of the potential). We always assume that W has no quadratic terms; then the category $\mathcal{C}$ has a spherical object for each vertex of Q, and W encodes the non-trivial products in a cyclic A_∞-structure on the total endomorphism algebra of those objects (this is the Koszul dual of the Ginzburg algebra). By a 'symplectic geometric model' for $\mathcal{C}(Q,W)$ we mean a configuration of Lagrangian spheres $\{L_v\}_{v\in \mathrm{Vert}(Q)} \subset (X,\omega)$ such that the number of edges between v and v' gives the rank of $HF^*(L_v,L_{v'})$ and the endomorphism algebra $\bigoplus_{v,v'} HF^*(L_v,L_{v'})$ is A_∞-quasi-isomorphic to that determined by the potential W. Such models are especially useful when the $\{L_v\}$ (split-)generate the compact category $\mathcal{F}(X,\omega)$. One can also seek algebro-geometric models for $\mathcal{C}(Q,W)$, i.e., embeddings $\mathcal{C}(Q,W) \hookrightarrow \mathcal{D}(Y)$ into the derived category of some quasi-projective Y (ideally with image some naturally characterised subcategory), or indeed algebraic models in terms of realisations in derived categories of modules over a non-commutative algebra. All of these will play a role in the sequel.

2. The double bubble and multi-bubbles

Recall from Section 1 that one can associate a 3-dimensional[3] Calabi–Yau category to a quiver with potential, i.e., a quiver equipped with a cyclic word on the edges. Arguably the simplest quiver admitting a non-trivial potential is the oriented 2-cycle:

$$\bullet \underset{f}{\overset{e}{\rightleftarrows}} \bullet, \qquad \text{potential } W = (ef)^2.$$

(Any potential on this quiver has the form $W = \sum_j a_j (ef)^j = (ef)^k(a_k + \cdots) \simeq (ef)^k$, up to formal change of variable and working over $\mathbb{C}$.) We write $\mathcal{C}$ for the corresponding CY_3-category.

A symplectic model X for $\mathcal{C}$, in the sense of Section 1, is given by the affine $(1,1)$-hypersurface in $\mathbb{P}^2 \times \mathbb{P}^2$. The symplectic completion of this affine variety is also described as the plumbing of two Lagrangian S^3s, say L_1, L_2, along an embedded unknotted circle S^1 such that the Morse–Bott Lagrangian surgery K of L_1, L_2 is again a 3-sphere. Considering suitable pencils and nets of hypersurfaces shows that there is a monodromy representation

$$\mathrm{PBr}_3 \to \pi_0 \,\mathrm{Symp}_{\mathrm{ct}}(X,\omega)$$

[2]When the arrows admit suitable gradings, one can associate a CY_n-category for any n.

[3]There is a generalisation to a 'graded quiver with potential' which allows one to associate a CY_d-category for other values of d, cf. [69].

with the pure braid group generated by the spherical twists τ_{L_1}, τ_{L_2}, τ_K; this can be seen more concretely from the realisation of X as a Morse–Bott–Lefschetz fibration over $\mathbb{C}$, which comes in a family of such where one moves the critical values of the fibration. Since the vanishing cycles differ at each singularity, only pure braids show up.

Theorem 2.1. *The natural monodromy map* $\mathrm{PBr}_3 \to \pi_0 \mathrm{Symp}_{\mathrm{ct}}(X, \omega)$ *splits.*

En route to showing this, it will be helpful to know the following.

Lemma 2.2. *Any compactly supported symplectomorphism of manifold* X *acts trivially on* $K_{\mathrm{num}}(\mathcal{F}(X)) \simeq \mathbb{Z}^2$.

Proof. The argument is very close to [45, Section 3.6]. The natural Euler pairing

$$K(\mathcal{F}(X)) \times K(\mathcal{W}(X)) \to \mathbb{Z}, \qquad (L, \Delta) \mapsto \chi(HF(L, \Delta))$$

descends to a non-degenerate pairing between $K_{\mathrm{num}}(\mathcal{F}(X))$ and $K(\mathcal{W}(X))$, where K_{num} was introduced in (1.1). The group $K(\mathcal{W}(X))$ is generated by the two cotangent fibres; one thus sees that $K_{\mathrm{num}}(\mathcal{F}(X)) \cong H_3(X; \mathbb{Z})$ is generated by the two compact core 3-spheres of the plumbing, and the non-degenerate pairing between $K_{\mathrm{num}}(\mathcal{F}(X))$ and $K(\mathcal{W}(X))$ is identified with the intersection pairing

$$H_3(X) \otimes H_3(X, \partial X) \to \mathbb{Z}. \tag{2.1}$$

The intersection pairing between the two core 3-spheres vanishes, indeed they are smoothly disjoinable (for a more striking and closely related statement see Corollary 6). Non-degeneracy of (2.1) then implies that $H_3(X) \to H_3(X, \partial X)$ vanishes. But then from the exact sequence

$$H_3(\partial X) \to H_3(X) \to H_3(X, \partial X)$$

which is functorial under symplectomorphisms, we see that compactly supported maps (which act trivially on the first term) act trivially on $H_3(X)$. □

There is also a geometric model for $\mathcal{C}$ in algebraic geometry. Take Y to be the crepant resolution $f\colon Y \to \mathrm{Spec}(R)$ of the isolated singularity given by $R = \mathbb{C}[[u, v, x, y]]/\langle uv - xy(x+y)\rangle$; then Y is a neighbourhood of two floppable $(-1,-1)$-curves meeting once. We consider the subcategory $\mathcal{C}_Y$ of the derived category of Y consisting of objects $\mathcal{E}$ supported on the exceptional locus and with $Rf_*\mathcal{E} = 0$ (this is split-generated by the $\mathcal{O}_{\mathbb{P}^1}(-1)$ for the two exceptional $\mathbb{P}^1$'s).

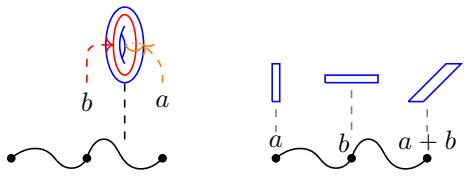


FIGURE 1. Vanishing cycles for Morse–Bott–Lefschetz fibration $X \to \mathbb{C}$, fibre $(\mathbb{C}^*)^2$.

There are actions of the pure braid group on X and on $D(Y)$ via symplectic parallel transport respectively via flop-flop functors. Mirror symmetry in this example states the following.

Theorem 2.3 ([84, Proposition 4.27 (1) and Corollary 5.7])**.** *There are equivalences*

$$\mathcal{F}(X) \simeq \mathcal{C}_Y \qquad \text{and} \qquad \mathcal{W}(X) \simeq D(Y)/\langle \mathcal{O}_Y \rangle$$

which entwine the action of the pure braid group.

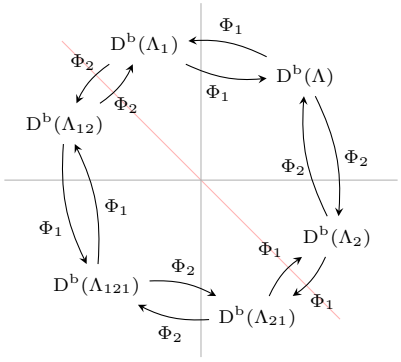
Remark 2.4. It seems likely that $D(Y)$ itself is a 'partially wrapped' category of X; however, it seems hard to characterise $D(Y)$ in a way which lets one pick out (and prove the equivalence to) a distinguished such. Recall that X is the complement of a generic $(1,1)$-divisor in the flag 3-fold, so X admits a projective compactification with smooth boundary divisor. This means there is a Reeb flow on ∂X which rotates the fibres of a circle bundle, and hence a cochain complex computing $SH^*(X)$ which has linear growth at chain level, compare to [75, Section 4a]. It follows that $SH^0(X)$ has Krull dimension at most 1, and $\operatorname{Spec} SH^0(X)$ has dimension at most 1. If $\mathcal{W}(X)$ was a categorical crepant resolution of the ring R in the sense of [65], one would have $\operatorname{Spec} SH^0(X) \cong \operatorname{Spec} R$ being 3-dimensional. Thus, our setting is somewhat different to the one often considered in mirror symmetry for log Calabi–Yau's, as discussed in [65].

The space of stability conditions on $\mathcal{C}_Y = \mathcal{C}$ was shown to be connected in [35], so the 'distinguished' component is preserved by all autoequivalences. They also proved that the 'Fourier–Mukai' equivalences (namely those induced from equivalences of $D(Y)$ commuting with Rf_*) define a group $\operatorname{Auteq}^{\mathrm{FM}}(\mathcal{C}) \simeq \mathrm{PBr}_3$. Unfortunately, the symplectic counterpart of commuting with Rf_* is not immediate. We circumvent this by using the classification of all hearts of bounded t-structures on $\mathcal{C}$.

We briefly recall some results on the Deligne groupoid that appears in this setting [21, 35]: this is a way of labelling a collection of 'mutation' equivalences between a finite set of derived categories associated to R by paths in a certain fundamental groupoid, with loops defining autoequivalences.

There is an explicit crepant resolution of $\operatorname{Spec}(R)$ associated to each ordering of the linear factors $\{\ell_1, \ell_2, \ell_3\} = \{x, y, (x+y)\}$ occurring in its defining equation, given by blowing up the ideals (u, ℓ_i) sequentially. The different resolutions arising from different orderings may be isomorphic, but are not canonically isomorphic. Thus, we obtain six different crepant resolutions.

One can place these six resolutions in the chambers of a hyperplane arrangement. General theory [84, Lemma 3.3] identifies these crepant resolutions with 'non-commutative crepant resolutions' (NCCRs), which are endomorphism algebras[4] of certain maximal Cohen–Macaulay modules over R. From this viewpoint, there are six NCCRs associated to the six geometric crepant resolutions, which are related by *mutation functors* Φ_1 and Φ_2, corresponding to mutation between the NCCRs of R at vertex 1, and vertex 2, respectively. See [84, Section 6a] for further explanation. These mutation functors form a representation of the Deligne groupoid associated to the given hyperplane arrangement, as follows:



[4] An explicit algebraic computation for one of the NCCRs, realising it as a quiver with relations [84, Lemma 3.5], pins down its A_∞-structure, which plays a key role in comparing it to the symplectic geometric models.

where the algebras Λ, Λ_1, Λ_2, Λ_{12}, Λ_{21}, Λ_{121} are the six NCCRs corresponding to the six crepant resolutions of $\mathrm{Spec}(R)$. As explained in [36, Remark 4.7], there is a functorial isomorphism $\Phi_1\Phi_2\Phi_1 \cong \Phi_2\Phi_1\Phi_2$.

There are simple modules $\mathcal{S}_1$ and $\mathcal{S}_2$ for Λ with the property that the A_∞-endomorphism algebra $\mathrm{Ext}^*_\Lambda(\mathcal{S}_1 \oplus \mathcal{S}_2)$ agrees with the Floer A_∞-algebra of the two core 3-spheres considered previously, cf. [84, Proposition 3.11]. By slight abuse of notation, we write $\mathcal{S}_1$, $\mathcal{S}_2$ for the simple modules for the NCCRs, regardless of which NCCR is under consideration. The labelling is consistent, so that mutation Φ_i always shifts $\mathcal{S}_i$.

Write C_+ for the top right chamber in the hyperplane arrangement.

Lemma 2.5. *There is a bijection between morphisms in the Deligne groupoid ending at C_+ and bounded t-structures on $\mathcal{C}$.*

Proof. This is the analogue of [5, Corollary 5.3] but dealing with bounded t-structures in place of tilting complexes. One considers the map $\alpha \mapsto \Phi_\alpha(\mathbb{T})$, with $\mathbb{T}$ the standard t-structure, as a map from morphisms in the groupoid to t-structures. Comparing to the proof in op. cit. the injectivity argument goes through exactly as written (using that the pure braid group action on $\mathcal{C}$ is faithful, by, e.g., [5, Corollary 5.5]), whilst the surjectivity argument follows from [35, Theorem 4.6]. □

Lemma 2.6. *If f is a compactly supported symplectomorphism of X inducing an autoequivalence F, there is a path α so that $F(S_i) = \Phi_\alpha(S_i)$ for all the simple objects in the standard heart.*

Proof. We know that F applied to the standard t-structure $\mathbb{T}$ with heart $\heartsuit$ yields some bounded t-structure $F(\mathbb{T})$ with heart $F(\heartsuit)$, so by the previously established bijection there is some α so that $F(\heartsuit) = \Phi_\alpha(\heartsuit)$. Now $F \circ \Phi_\alpha^{-1}$ fixes $\heartsuit$ and hence permutes the simples. By Lemma 2.2, F acts trivially on K-theory so induces the identity permutation. On the other hand, the combinatorics of [36, Proposition 4.8] shows that Φ_α can only act on K-theory via one of the six possible matrices realised by the compositions of the standard reflection functors, and one checks directly that only the identity map acts by a permutation matrix. The result follows. □

Lemma 2.7. *If $\Phi_i^\pm$ denote the left and right tilts at the simple S_i, then $F \circ \Phi_i^\pm$ and $\Phi_\alpha \circ \Phi_i^\pm$ send the standard heart to the same heart.*

Proof. The definition of the right and left tilts depends only on the heart and choice of simple. The result then follows from Lemma 2.6, which implies that $F \circ \Phi_i^\pm$ and $\Phi_\alpha \circ \Phi_i^\pm$ act in the same way on the torsion pair defining the tilt. □

Lemma 2.8. *All bounded hearts in $\mathcal{C}$ are obtained from a sequence of tilts starting from the standard heart (i.e., the heart exchange graph is connected).*

Proof. By Lemma 2.5, every bounded t-structure is of the form $\Phi_\alpha(\mathbb{T})$, where Φ_α is a composition of mutation functors and their inverses. The statement follows using the fact that the mutation functors (and their inverses) give the left and right tilts at simples (see, e.g., [36, Lemma 5.5]). □

We now induct over the number of tilts needed to reach some given heart from the standard heart. This is a 'careless' induction, in the sense that we are not appealing to any normal form for such a sequence of tilts: we are just exhausting all hearts by arbitrary sequences of tilts starting from the standard heart. By this induction, given the α from Lemma 2.6 we conclude that F and Φ_α agree on the t-structure obtained from $\heartsuit$ by applying any element of the braid group, i.e.,

$$F\Phi_\gamma(\heartsuit) = \Phi_\alpha(\Phi_\gamma(\heartsuit))$$

for every path γ in the Deligne groupoid ending in the chamber C_+. Using the bijection from Lemma 2.5, we see that F and Φ_α agree on all bounded hearts. The following now completes the proof of Theorem 2.1.

Corollary 2.9. *The association $f \mapsto F$ of Lemma 2.6 defines a homomorphism*

$$\pi_0 \operatorname{Symp}_{\mathrm{ct}}(X) \to \mathrm{PBr}_3, \qquad f \mapsto \alpha, \qquad \text{where} \quad F(\heartsuit) = \Phi_\alpha(\heartsuit).$$

Proof. Given symplectomorphisms f, g yielding autoequivalences F, G which act trivially on K-theory, we define pure braids α and β by the property that

$$F(\heartsuit) = \Phi_\alpha(\heartsuit), \qquad G(\heartsuit) = \Phi_\beta(\heartsuit).$$

So then $FG(\heartsuit) = F(\Phi_\beta(\heartsuit))$ and we wish to show this equals $\Phi_\alpha\Phi_\beta(\heartsuit)$. But this follows immediately from the discussion above, which established that F and Φ_α act the same way on all bounded hearts. □

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